

DESIGN AND IMPLEMENTATION OF PRESSURE FEEDBACK FOR LOAD-CARRYING APPLICATIONS WITH POSITION CONTROL

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ABSTRACT

This research paper presents the design and implementation of pressure feedback on a hydraulically actuated single-boom crane operated in closed-loop position control. It is well known that systems with pressure compensated proportional valves in combination with over-center valves tend to induce instability, especially when the external load is overrunning (e.g., while lowering a load). However, in some applications pressure oscillations arise also with resistant external loads (e.g., while lifting a load). Hence, a pressure feedback capable of stabilizing the system functioning in both operations using the pressures from both actuator chambers (i.e., piston-side and rod-side) is proposed and compared against the conventional solution using only the rod-side chamber pressure. The investigation demonstrates that the implementation of a proposed “inverse valve dynamic” algorithm is needed in order for the control valve to stabilize the system when introducing the piston-side pressure in the pressure feedback. With this new method, the experimental tests demonstrate a satisfactory reduction of the pressure fluctuations in closed-loop motion control and a good position tracking (the average position error while lowering the load is reduced by almost 90% compared to the original system without pressure feedback). Finally, simulated results show that the proposed pressure feedback allows for potential energy savings of about 50% when lowering the load.

KEYWORDS: Over-center valves, pressure compensated proportional valves, active-damping, pressure feedback, position control, energy saving.

1. INTRODUCTION

The hydraulic circuits of load-carrying applications must contain passive load-holding devices to meet safety regulations. A popular load-carrying solution makes use of pressure-compensated Proportional Directional Control Valves (PDCVs) in combination with Over-Center Valves (OCVs). This is considered state-of-the-art in many industrial fields (e.g., in knuckle-boom cranes for offshore applications) due to the load-independent flow control and reduced system cost. Nevertheless, this approach tends to introduce an oscillatory behavior, or even instability, that undermines both performance and operational safety. Several investigations were carried out to mitigate this issue, especially when dealing with overrunning loads. Optimizing the design parameters [1] is today a common approach that often requires hardware reconfiguration [2] (e.g., adjusting the pilot ratio of the OCV's, increasing the volumes in hydraulic lines related to the OCV pilot pressure and/or the load-sensing pilot pressure of the PDCV's compensator). An alternative is introducing active damping to compensate for oscillations. An overview of active damping focused on hydraulically actuated mobile applications is presented in [3]. This solution is usually referred to as Pressure Feedback (PF) and may involve closed-loop control of the PDCV's spool position [4]. In [5], a high-pass filter is implemented in the PF of a

mobile-crane and demonstrates a clear reduction of the oscillatory behavior. More advanced control strategies using PF are proposed in [6-8] (i.e., adaptive control, frequency-based control and auto tuning). However, according to [9], these methods are complex and less general compared to the approach first described in [4], referred to as the “classical approach”. Recently, Pedersen and Andersen demonstrated that there is an optimum range for the high-pass filter parameters that affects both the obtainable damping and the dynamic response [9]. A guideline for adjusting the filter parameters is proposed in [10] for systems with pressure-compensated PDCVs in combination with OCVs. However, the low bandwidth of PDCVs may have essential limitations on the damping effect [9-11], especially when using PDCVs that have a much lower bandwidth compared to servo-valves. Moreover, pressure feedback is typically implemented in combination with open-loop velocity control where the machine operator closes the loop (operator in the control loop). However, in some applications (e.g., automated offshore oil drilling machinery [12]) a fully automated closed-loop position control is applied for more efficient operations. In the offshore environment, these machines are subjected to disturbances such as strong wind and relevant wave motion that may excite resonance modes of the mechanical structure and restrict the dynamic performance of the actuation system. These factors, when combined with the intrinsic oscillatory behavior of the hydraulic system reduce the system productivity. Consequently, there is an on-going trend of replacing hydraulic actuators with electric drives that results in increased energy efficiency and controllability [12-13].

For these reasons, this paper aims to design and implement pressure feedback on a standard valve-controlled actuator equipped with an over-center valve. The main focus is on a pressure feedback algorithm that uses both actuator pressures instead of only the pressure of the rod-side chamber, like [5] and [10-11]; this idea is inspired by a solution successfully implemented on a self-contained electro-hydraulic cylinder with passive load-holding capability that drives a single-boom crane in two quadrant operations [14]. The considered system under investigation is explained in detail in Section 2. A non-linear model is described and validated against experimental data in Section 3.1, and further simplified and linearized in Section 3.2. The linear model is in Section 4 used to analyze the stability of the original system and when implementing the considered pressure feedback methods. In Section 5, the proposed algorithm with pressure feedback is tested on a high-fidelity non-linear model before the final implementation on the experimental test-bed. The last section contains the conclusion.

2. THE CONSIDERED SYSTEM

The considered system is the hydraulically-actuated, single-boom crane visible in Figure 2 that is built specifically for having a flexible structure in the boom with the purpose of inducing pressure oscillations in the hydraulic system depicted in Figure 1. The OCV serves multi-functional purposes such as leak tight load-holding and shock absorption [11]. A state-of-the-art pressure compensated PDCV is connected to a centralized Hydraulic Power Unit (HPU) providing a constant supply pressure (p_s) and a fixed return pressure (p_R) to the actuation system. The motion of the hydraulic cylinder (C) is controlled by the PDCV that receives the control input u_v from the control system.

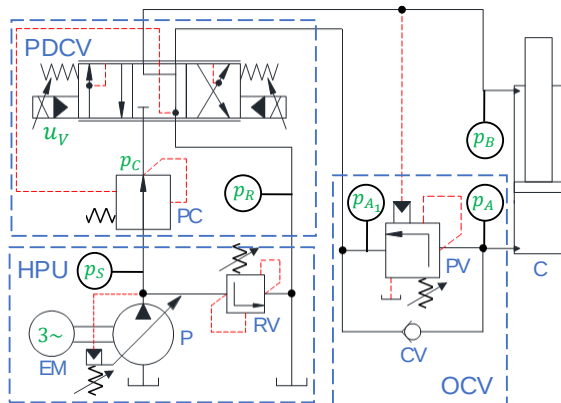


Figure 1. The hydraulic system of the experimental setup.

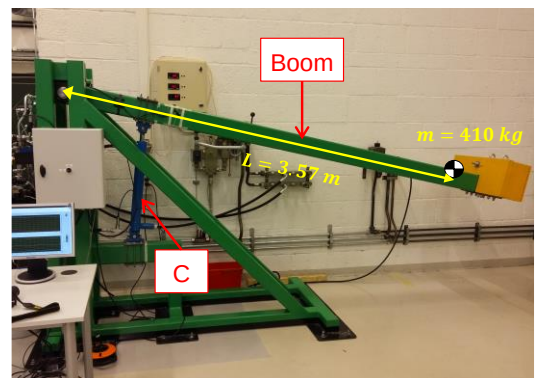


Figure 2. The single-boom crane.

The key components for the experimental setup consist of the following eight elements in addition to the instrumentation: the electric motor (EM) drives the variable-displacement, axial-piston pump (P) with displacement of $75 \text{ cm}^3/\text{rev}$. The supply pressure is controlled by the absolute pressure limiter of the pump while a pressure-Relief Valve (RV) is installed for safety. The PDCV, a Danfoss PVG32, consists of two key parts, namely the main spool with integrated closed-loop position control and the Pressure Compensator (PC) that guarantees a load independent flow across the metering edge. The vented OCV, a Sun Hydraulics CWCA with pilot ratio 3:1, contains a by-pass Check Valve (CV) and a pilot-operated Poppet Valve (PV) for controlling the overrunning load when lowering the crane boom. The hydraulic cylinder has piston diameter $D_p = 65 \text{ mm}$, rod diameter $D_R = 35 \text{ mm}$, and stroke length $h_c = 500 \text{ mm}$. Finally, the system is instrumented with sensors for measuring the pressures labeled in Figure 1 as well as the cylinder position (x_c).

3. SYSTEM MODELING AND VALIDATION

This paper makes use of a dynamic model of the valve-controlled system that drives the single-boom crane: the MATLAB-Simulink® environment was chosen to perform the numerical simulations. In this section both a high-fidelity, non-linear model and a simplified linear model are introduced.

3.1. The High-fidelity Model

The mechanical system including the crane boom is modeled using the finite segment method [15-16] (Figure 3), which serves the purpose of representing the relevant flexibility of the structure.

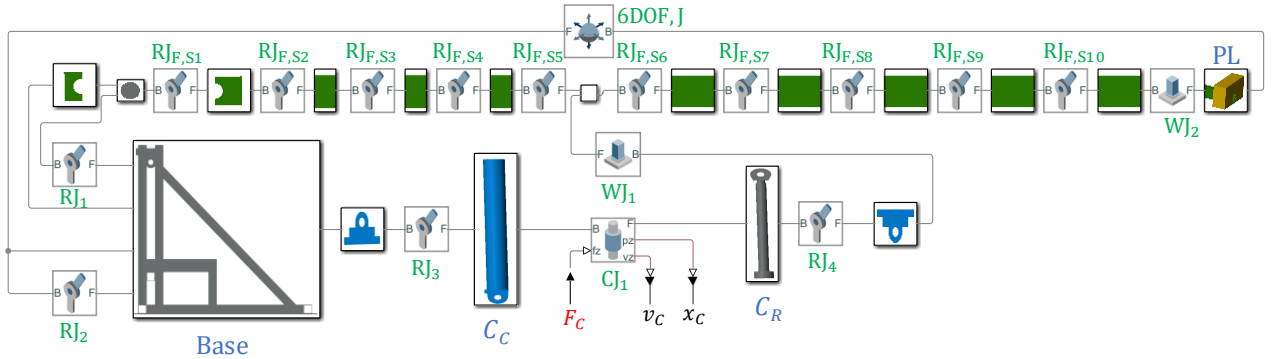


Figure 3. MATLAB-Simulink® multibody model of the flexible boom.

The multibody system (MBS) consists of two revolute joints (RJ_{1-2}) connecting the flexible boom to the base of the crane and two RJ_{3-4} connecting the cylinder body (C_C) to the base and the rod (C_R) to the boom, respectively. A cylindrical joint (CJ) is applied between C_C and C_R interfacing the hydraulic system through the cylinder piston force (F_C), velocity (v_C) and position (x_C). In the finite segment method, the boom is modeled by a number of smaller boom segments connected together using RJs with an internal torsional spring/damper ($RJ_{F,S1-10}$) to simulate the bending of the structure. The torsional springs have equivalent stiffness ($k_{eq,i}$) and damping ($C_{eq,i}$) related to the segment number i , as expressed in (1) and (2), where E is the bulk modulus of the boom material, I_z is the second moment of inertia for the cross section of the boom, $L_{S,i}$ is the length of the segment, and $J_{S,i}$ is the moment of inertia of the segment.

$$k_{eq,i} = \frac{EI_z}{L_{S,i}} \quad (1)$$

$$C_{eq,i} = \sqrt{2J_{S,i}k_{eq,i}} \quad (2)$$

Concerning the hydraulics, equation (3) elucidates the actuator force transferred to the MBS.

$$F_c = p_A A_A - p_B A_B - F_f \quad (3)$$

The actuator pressures (p_A and p_B) and areas (A_A for the piston-side and A_B for the rod-side) describe the hydraulic force while the friction force (F_f) is given in (4), according to the Stribeck model.

$$F_f = B_{vf} \dot{x} + \tanh(\dot{x}a) \left(F_{cf} + F_{sf} e^{-\frac{\dot{x} \cdot \tanh(\dot{x}a)}{\tau_{sf}}} \right) \quad (4)$$

The different terms identified with experiments are the viscous friction coefficient ($B_{vf} = 1.5 \cdot 10^4$ kg/s), the Coulomb force ($F_{cf} = 75$ N), the static friction force ($F_{sf} = 1450$ N), and the static friction force's constant ($\tau_{sf} = 0.02$ m/s). The hyperbolic tangent is introduced to prevent numerical issues, where the dimensionless tuning parameter is set as $a = 250$.

The well-known pressure build-up equation is then applied to evaluate the pressures labeled in Figure 1 (both the supply pressure $p_s = 180$ bar and the return pressure $p_R = 1.1$ bar are assumed constant). The resulting expressions are given in (5)-(7).

$$\dot{p}_{A1} = \frac{1}{C_{A1}} (Q_{V,in} - Q_{CV} + Q_{PV}), \quad \text{where } C_{A1} = \frac{V_{A1,0}}{\beta_{A1}} \quad (5)$$

$$\dot{p}_A = \frac{1}{C_A} (Q_{CV} - A_A v_C - Q_{PV}), \quad \text{where } C_A = \frac{A_A x_C + V_{A,0}}{\beta_A} \quad (6)$$

$$\dot{p}_B = \frac{1}{C_B} (Q_{V,out} + A_B v_C), \quad \text{where } C_B = \frac{A_B (h_C - x_C) + V_{B,0}}{\beta_B} \quad (7)$$

The hydraulic capacitances related to the actuator chambers (C_i) are defined recalling the piston position when needed, the cylinder stroke, the volumes of the transmission lines ($V_{i,0}$) that are assumed constant, and the effective fluid's bulk modulus (β_i). This parameter is modeled differently for each capacitance via equation (8).

$$\beta_i = \frac{1}{\frac{1}{\beta_0} - \frac{1}{\kappa_{air}(p_i + p_{atm}) \left(\frac{\eta_{air} - 1}{\eta_{air} \left(\frac{p_{atm}}{p_i + p_{atm}} \right)^{\frac{1}{\kappa_{air}}} - 1} \right)}} \quad (8)$$

The terms are the oil's bulk modulus ($\beta_0 = 12000$ bar), the adiabatic air constant ($\kappa_{air} = 1.4$), the pressure in the capacitance (p_i), the atmospheric pressure (p_{atm}) and the volumetric air content of the oil ($\eta_{air} = 0.007$ %). Additionally, the flow rates through the OCV are computed using a simplified approach (9) that involves the valve flow gain ($k_{V,i}$), the cracking pressure ($p_{Cr,i}$), the inlet pressure ($p_{In,i}$), and the outlet pressure ($p_{Out,i}$).

$$Q_i = \begin{cases} 0, & p_{In,i} < p_{Out,i} + p_{Cr,i} \\ k_{V,i}(p_{In,i} - p_{Out,i} - p_{Cr,i}), & p_{In,i} \geq p_{Out,i} + p_{Cr,i} \end{cases} \quad (9)$$

On the contrary, the flow rates through the PDCV are modeled using the orifice equation (10).

$$Q_i = \xi_v C_d A_d \sqrt{\frac{2}{\rho} |\Delta p_i|} \text{sign}(\Delta p_i) \quad (10)$$

It comprises the valve opening (ξ_v) (i.e., a dimensionless number ranging between -1 and 1 with zero being the center position of the spool), the discharge coefficient (C_d), the maximum flow area (A_d), the fluid density (ρ), and the pressure differential across the spool (Δp_i). The valve opening characteristics, such as the maximum flow areas as well as the spool overlaps (dead-bands) are taken from [17]. The spool position is related to the valve command (u_v) via a second order transfer function ($G_v(s)$ in (11)) that accounts for the valve dynamics. The natural frequency is $\omega_v = 30 \frac{\text{rad}}{\text{s}}$ as identified in [18] while the damping ratio is adjusted equal to $\zeta_v = 0.7$ after validation against experimental data.

$$G_V(s) = \frac{\xi_V}{u_V} = \frac{\omega_V^2}{s^2 + 2\omega_V\zeta_V s + \omega_V^2} \quad (11)$$

The PDCV's upstream pressure (p_c) is modeled according to (12); the nominal pressure drop across the spool's metering edge (i.e., the equivalent pressure setting of the compensator $p_0 = 7 \text{ bar}$) and the load-sensing pressure (p_{LS}) are introduced. The latter term is selected by a logic function so that $p_{LS} = p_A$ when the spool position is negative, $p_{LS} = p_B$ when the spool position is positive, and $p_{LS} = p_R$ when the spool is centered.

$$p_c = \begin{cases} p_{LS} + p_0, & p_0 \leq (p_S - p_{LS}) \\ p_S - p_{LS}, & 0 < (p_S - p_{LS}) < p_0 \\ p_{LS}, & (p_S - p_{LS}) \leq 0 \end{cases} \quad (12)$$

The relative opening of the poppet (ξ_{PV}) in the OCV is given in (13), where α_p is the pilot area ratio, $p_{PV,cr}$ is the cracking pressure, and $p_{PV,fo}$ is the extra pressure required to fully open the valve.

$$\xi_{PV} = \frac{\alpha_p \cdot p_B + p_A - p_{PV,cr}}{p_{PV,fo}} \quad (13)$$

The dynamics of the poppet and the dynamics of the pressure compensator are simulated via a first-order transfer function with a time constant $\tau_i = 0.001 \text{ s}$.

Finally, validation against experimental data was carried out and the results are reported in Figure 4. The comparison shows a good match (except the pressure peak that occurs when the PDCV closes after 20s in Figure 4 (d)) between simulated and measured data, especially, with regards to the oscillatory behavior.

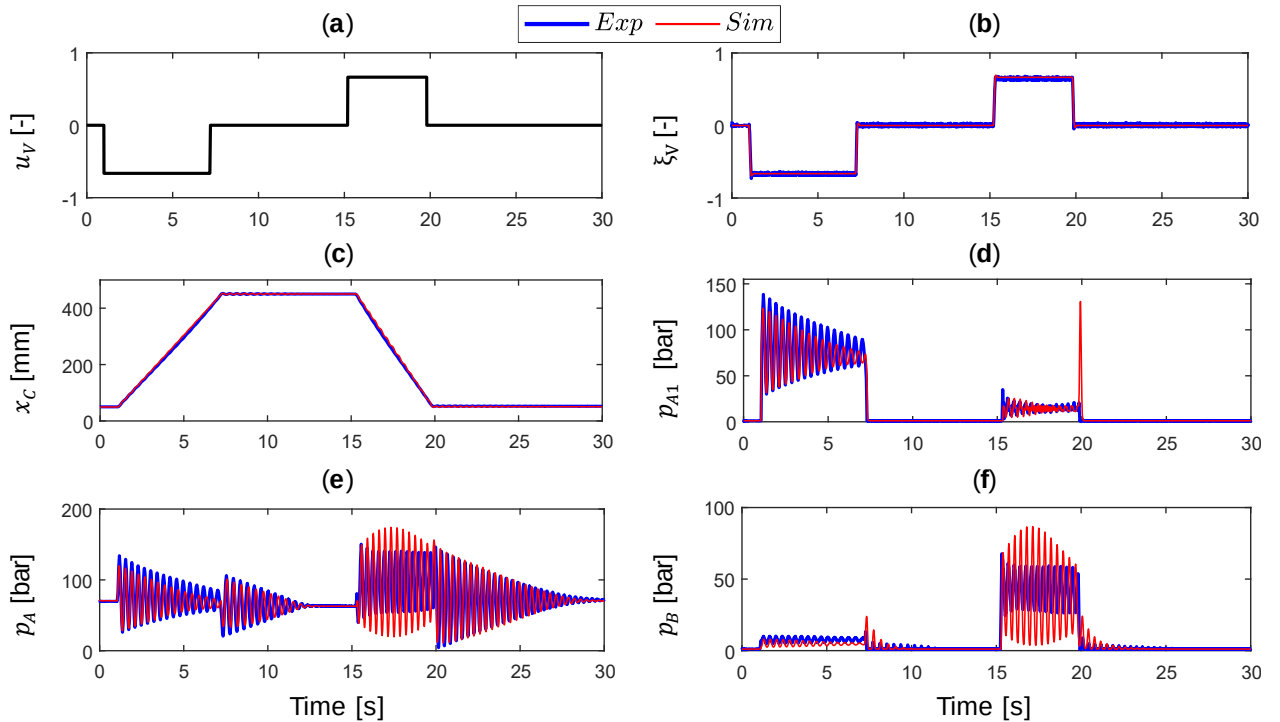


Figure 4. Validation of the non-linear model against experimental data: (a) Input command; (b) PDCV spool position; (c) Cylinder position; (d)-(f) Relevant system pressures.

3.2. The Linear Model

A simplified model emphasizing the cylinder retraction (i.e., the most critical operating condition) was derived for a linear analysis. The flow through the CV elucidated in (5)-(6) is neglected and all capacitances are considered constant and defined by the cylinder position $x_{C,0} = 0.45 \text{ m}$. The resulting force balance on the piston is given in (14).

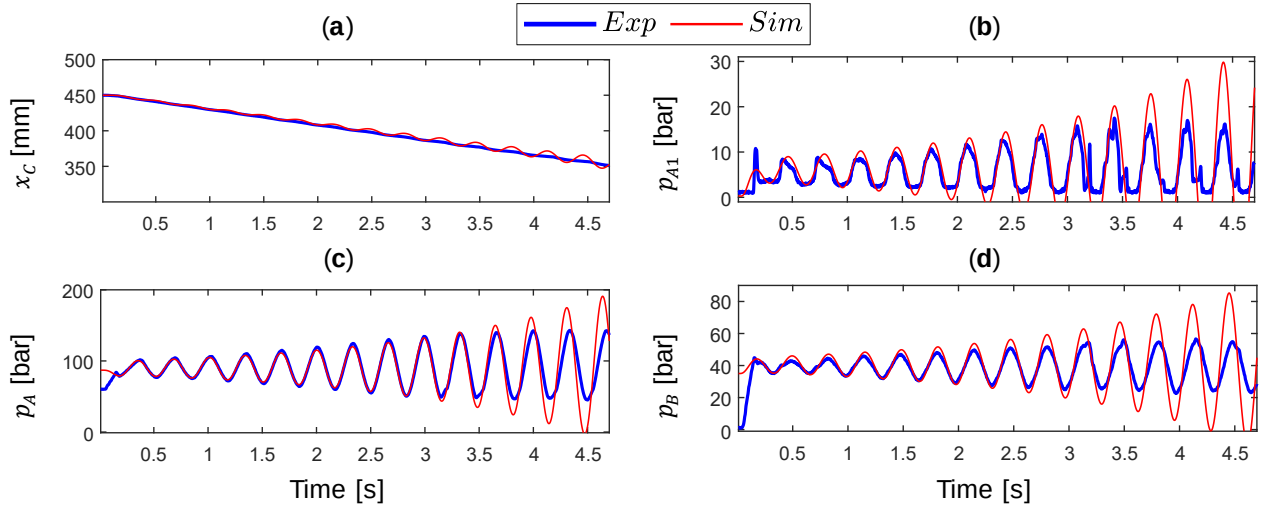


Figure 6. Validation of the linear model against experimental data: (a) Cylinder position; (b)-(d) System pressures.

4. PRESSURE FEEDBACK

Implementing high-pass filtered pressure feedback is proven to increase the system damping for cranes when lowering the load [5]. A first-order high-pass filter (16) is chosen.

$$G_f(s) = \frac{Q_{PF}}{p_i} = \frac{k_f s}{\tau_f s + 1} \quad (16)$$

The two following methods of PF are investigated in this paper.

4.1. The Rod-side Pressure Feedback

First, the rod-side pressure (p_B) is used as feedback element (PF_{p_B}) in accordance to the approach chosen in [5], [10] and [11] as shown in Figure 7.

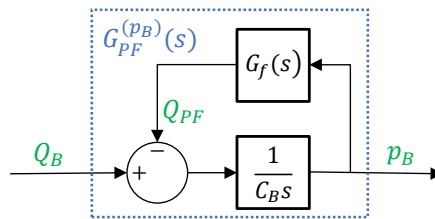


Figure 7. Block diagram of rod-side pressure feedback.

By using the high-passed pressure feedback, the open-loop transfer function $G_{p_B}(s)$ of the original system (Figure 5) becomes the compensated open-loop transfer function $G_{PF}^{(p_B)}(s)$ according to (17).

$$G_{PF}^{(p_B)}(s) = \frac{1}{(C_B + k_f)s} \frac{\tau_f s + 1}{\frac{C_B}{C_B + k_f} \tau_f s + 1} \quad (17)$$

For the implementation of this PF on a real application (Figure 8), the non-negligible dynamics of the PDCV should be accounted for when selecting the filter parameters. The inverse of the valve flow gain ($k_{q,Valv}$) is, therefore, included to estimate the required valve opening.

$$G_{x_C}^{(p_B)}(s) = \frac{x_C}{u_v}_{(ref)} = \mathbf{C}_{x_C} (\mathbf{I}s - \mathbf{A}^{(p_B)})^{-1} \mathbf{B}, \quad \text{where } \mathbf{C}_{x_C} = [1 \ 0 \ 0 \ 0 \ 0] \quad (20)$$

Figure 9 (a) shows a clear increase in damping when PF is added. Figure 9 (b) suggests that the system without PF is stable when the cylinder position is below 0.313m. Figure 9 (c) displays that the two positive complex poles of PF_{off} move to the negative side when adding pressure feedback (PF_{pB}) in the worst-case scenario (i.e., $x_c = 0.45$ m), resulting therefore in a stable system.

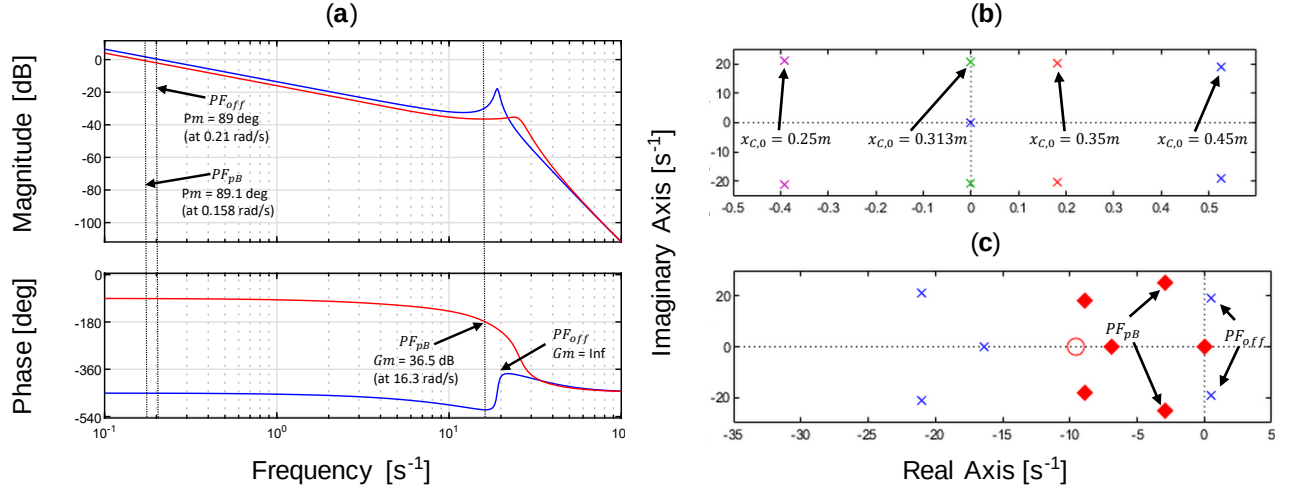


Figure 9. Stability analysis of $G_{x_c}^{(p_B)}(s)$: (a) Bode plots; (b) Plot PF_{off} at different cylinder positions; (c) Poles and zeros PF_{off} (blue) and PF_{pB} (red) at $x_{c,0} = 0.45$ m.

4.2. The Load Pressure Feedback

A different approach for the pressure feedback based on the “load pressure” (i.e., the difference between the bore chamber pressure multiplied by the area ratio of the cylinder and the rod chamber pressure, $p_L = p_A \mu_c - p_B$) is addressed in order to get rid of the undesired pressure oscillations identified during piston extension (Figure 12). Achieving this result benefits those systems characterized by a flexible mechanical structure like the single-boom crane under investigation.

The row eight of matrix A , in (18) is modified according to (21) when including the piston-side chamber pressure.

$$A^{(p_L)} = \begin{bmatrix} \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \frac{-k_f A_B \left(\frac{\mu_c^2 + 1}{C_A C_B} \right)}{\tau_f} & 0 & \frac{-k_f \mu_c k_{q,PV} k_{qA,RV}}{C_A \tau_f} & \frac{-k_f \mu_c k_{q,PV} k_{qB,PV}}{C_B \tau_f} & \frac{-k_{qV,In} k_f}{C_B \tau_f} & 0 & \frac{-1}{\tau_f} \end{bmatrix} \quad (21)$$

The open-loop transfer function from valve command to position output ($G_{x_c}^{(p_L)}(s)$) elucidated (22) and the transfer function from valve command to velocity output ($G_{v_c}^{(p_L)}(s)$) elucidated (23) are used to analyze the system stability using load pressure feedback (PF_{pL} , Figure 10). The filter's time constant $\tau_f = \frac{2}{\omega_n}$ and the filter gain tuning parameter $\lambda_f = 0.7$ are used.

$$G_{x_c}^{(p_L)}(s) = \frac{x_c}{u_{(ref)}} = C_{x_c} (I s - A^{(p_L)})^{-1} B, \quad \text{where } C_{x_c} = [1 \ 0 \ 0 \ 0 \ 0] \quad (22)$$

$$G_{v_c}^{(p_L)}(s) = \frac{v_c}{u_{(ref)}} = C_{v_c} (I s - A^{(p_L)})^{-1} B, \quad \text{where } C_{v_c} = [0 \ 1 \ 0 \ 0 \ 0] \quad (23)$$

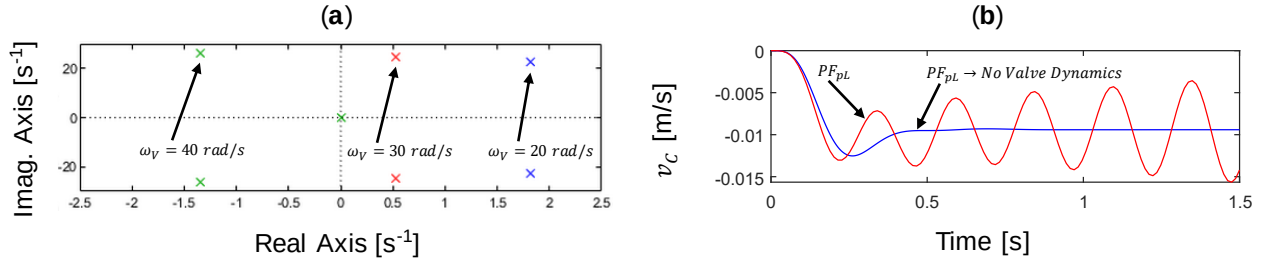


Figure 10. Stability analysis $x_{c,0} = 0.45m$: (a) Pole-zero plot of $G_{x_C}^{(p_L)}(s)$; (b) Step response of $G_{v_C}^{(p_L)}(s)$.

Figure 10 (a) shows that the system with PF_{pL} is unstable with the considered valve ($\omega_v = 30 \text{ rad/s}$); however, increasing the natural eigenfrequency of the valve (e.g., up to $\omega_v = 40 \text{ rad/s}$) makes the system stable. Figure 10 (b) compares a step response of $G_{v_C}^{(p_L)}$ with command $u_v = 10\%$ to highlight that the steady-state piston velocity is not achieved when considering the valve dynamics. The latter scenario, as discussed in [9-11], is because of the low bandwidth of PDCVs, and in [9] it is seen that when $\omega_v \geq 3\omega_n$ the effect diminishes. Consequently, the inverse of the valve dynamics ($G_v^{-1}(s)$) must be included in the pressure feedback according to the block diagram in Figure 11.

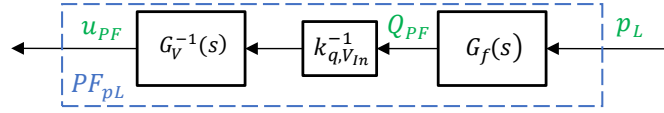


Figure 11. Pressure feedback using the load pressure and including the inverse valve dynamics.

To get a proper transfer function (i.e., the degree of the numerator does not exceed the degree of the denominator), a pseudo transfer function (24) is added assuming $k = 5$.

$$G_v^{-1}(s) = \frac{k^2 s^2 + 2\omega_v \zeta_v k^2 s + \omega_v^2 k^2}{s^2 + 2\omega_v k s + \omega_{n,v}^2 k^2} \quad (24)$$

5. RESULTS

First, the high-fidelity model is used to test the PF based on the rod-side pressure when compared against the original system. Secondly, the load pressure feedback is also evaluated before implementation and testing on the experimental setup. Finally, the proposed pressure feedback is simulated for different OCV's pilot ratios to understand if there is room for potential energy savings.

5.1. The Rod-side Pressure Feedback

To evaluate the performance of this pressure feedback, positive and negative step inputs at 40% of the maximum valve opening are simulated by means of the non-linear dynamic model (Figure 12).

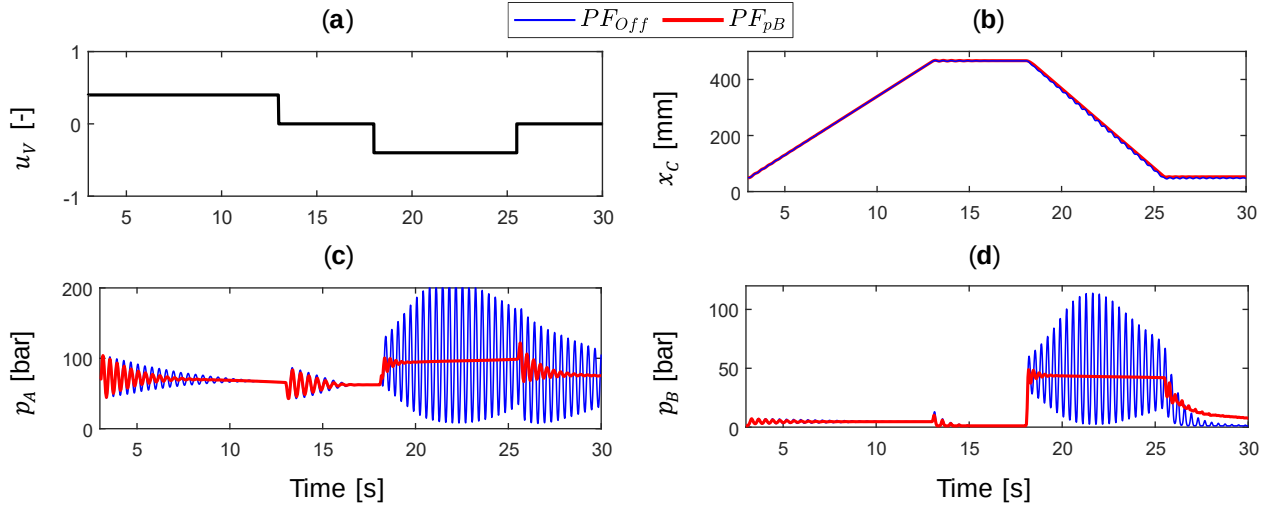


Figure 12. Simulated results with (PF_{pB}) and without (PF_{off}) pressure feedback: (a) System input; (b) Piston position; (c) Pressure in the bore chamber; (d) Pressure in the rod chamber.

The results in Figure 12 demonstrate a significant improvement when pressure feedback is used since the pressure oscillations are greatly reduced in comparison to the undamped system. This is especially true during piston retraction (i.e., from about 17 to 26 seconds) with an overrunning external load. However, when the cylinder is extending, the pressure feedback based on the rod chamber pressure is not able to mitigate the pressure oscillations (i.e., before 5 seconds). For this reason, the following feedback method involving both actuator pressures is investigated.

5.2. The Load Pressure Feedback

The performance of the proposed pressure feedback (PF_{pL}) is compared in Figure 13 against the conventional feedback approach (PF_{pB}) while simulating a step input.

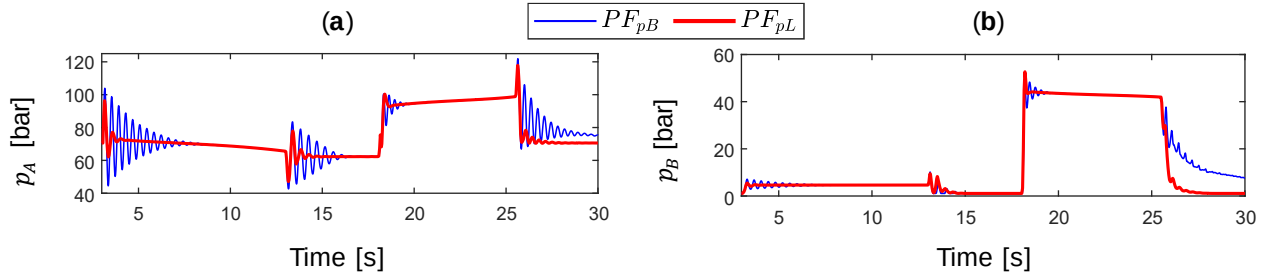


Figure 13. Simulated results of pressure feedback based on rod-side pressure and on load pressure when including $G_v^{-1}(s)$: (a) Pressure in the bore chamber; (b) Pressure in the rod chamber.

The results show a clear improvement of the load pressure feedback in reducing the unwanted pressure oscillations when the piston is extending (i.e., from 0 up to about 12 seconds). In addition, the performance is also improved during piston retraction.

5.3. Experimental Tests – The Closed Loop Position Control

The investigated load pressure feedback is tested experimentally on the considered application and compared to the conventional system that is used as a reference after selecting the filter parameters properly. Figure 14 illustrates the block diagram representing the implemented control algorithm.

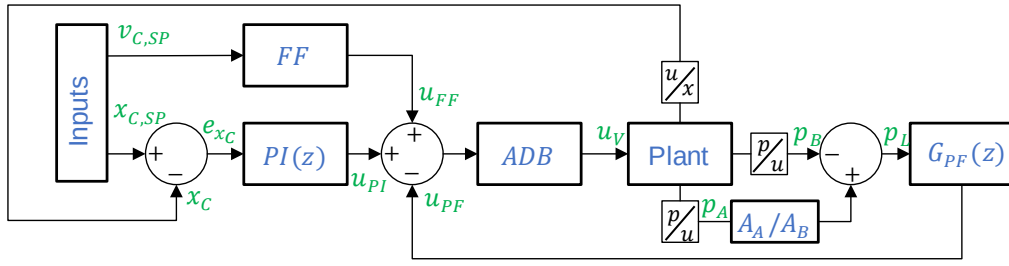


Figure 14. Block diagram of the considered control algorithm.

It consists of a position controller based on a closed-loop PI contribution (u_{PI}) and a feed forward (FF) term elucidated in (25), in addition to the pressure feedback in (26). Anti-dead-band logic is also implemented to compensate for the effective dead-band of the PDCV, as explained in [17].

$$u_{FF} = \frac{A_A v_C}{k_{Q_{max}}} \text{ if } v_C \geq 0; \quad u_{FF} = \frac{A_B v_C}{k_{Q_{max}}} \text{ otherwise} \quad (25)$$

$$G_{PF}(z) = \frac{u_{PF}}{p_L} = G_V^{-1}(z) k_{q, v_{in}}^{-1} G_f(z) \quad (26)$$

This controller was first developed in a Simulink® model and then converted to IEC 61131-3 Structured Text code using the Simulink-PLC-Coder®. The generated PLCopen XML file was then uploaded to IndraWorks® and executed on an embedded controller XM22 from Bosch Rexroth that runs at a frequency of 1000 Hz. The Simulink-PLC-Coder® requires discrete blocks, hence all transfer functions in the s-domain were converted to the z-domain.

A setpoint S-curve trajectory with a resulting piston velocity $v_{C,SP} = 20 \text{ mm/s}$ was generated and used as input to the position controller (a low velocity is assumed since it is the worst condition due to reduced damping). The measured position error for three different approaches are compared in Figure 15, namely a controller without PF, a control algorithm with PF based on p_B , and a control logic with PF grounded on p_L .

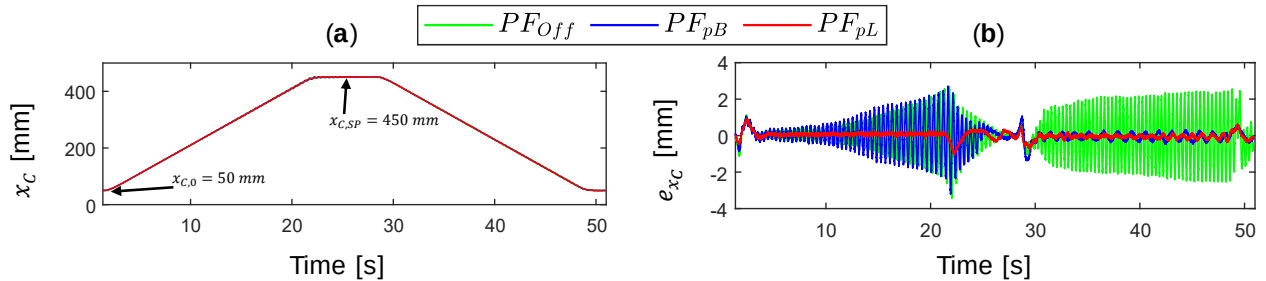


Figure 15. Experimental closed-loop position tracking for different control algorithms.

These experiments demonstrate that adding pressure feedback plays a major role in improving the system performance, even though the induced oscillations make the system with rod-side pressure feedback (PF_{pB}) unstable when the cylinder is extending. Conversely, the system with load pressure feedback (PF_{pL}) stabilizes the system during both piston extension and retraction. This approach results in a much smaller average position error during piston extension (the maximum error is 1.0 mm at about 22 s when the system is decelerating); in comparison to the system without pressure feedback, the improvement in terms of position tracking is close to 2 mm during piston retraction.

The smoother system behavior is clearly reflected in the pressures trends proposed in Figure 16: implementing load pressure feedback removes the oscillatory nature in both quadrants when closed-loop position control is desired.

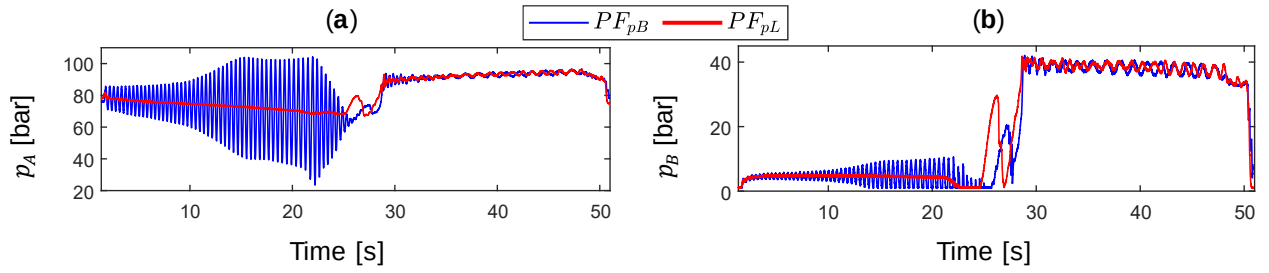


Figure 16. Experimental results comparing PF_{pL} and PF_{pB} : (a) Piston-side pressure; (b) Rod-side pressure.

A similar test for a higher piston velocity ($v_{c,SP} = 120 \text{ mm/s}$) that represents the upper limit in terms of velocity requirement for this system is also carried out showing similar behavior (the results are not reported for the sake of brevity).

Lastly, Table 2 provides a synthesis of the system performance for closed-loop position control: the simpler implementation of the rod-side pressure feedback is highlighted (i.e., including the valve dynamics is not strictly necessary) as well as the improved performance of the load pressure feedback (i.e., smoother functioning is achieved in every operation).

Table 2. Performance summary.

Pressure Feedback	$G_V^{-1}(s)$	Lifting	Lowering
Off		-	-
p_B	No	-	+
p_L	Yes	+	+

5.4. The Energy Saving Potential

The OCV's pilot ratio affects both the system stability and the energy efficiency. From the stability criterion given in (19), it is clear that it is not possible to achieve stability for the whole cylinder stroke when standard values of the pilot ratio (i.e., $\alpha_p = 2 \dots 8$) are used [11]. However, implementing pressure feedback can make the system stable. Solving this issue, provides an opportunity to emphasize the system's energy consumption that can potentially be improved by selecting a higher pilot ratio of the OCV. Energy savings can be achieved when lowering the crane if the pressure required to open the OCV is reduced; Figure 17 shows the effects of replacing the reference pilot ratio ($\alpha_p = 3$) with $\alpha_p = 5$ and $\alpha_p = 10$, respectively. The results are derived via the non-linear model simulated in closed-loop position control; the considered scenario is similar to the one discussed in Figure 16 but for a higher piston velocity ($v_{c,SP} = 120 \text{ mm/s}$) during piston retraction (i.e., when the OCV regulates).

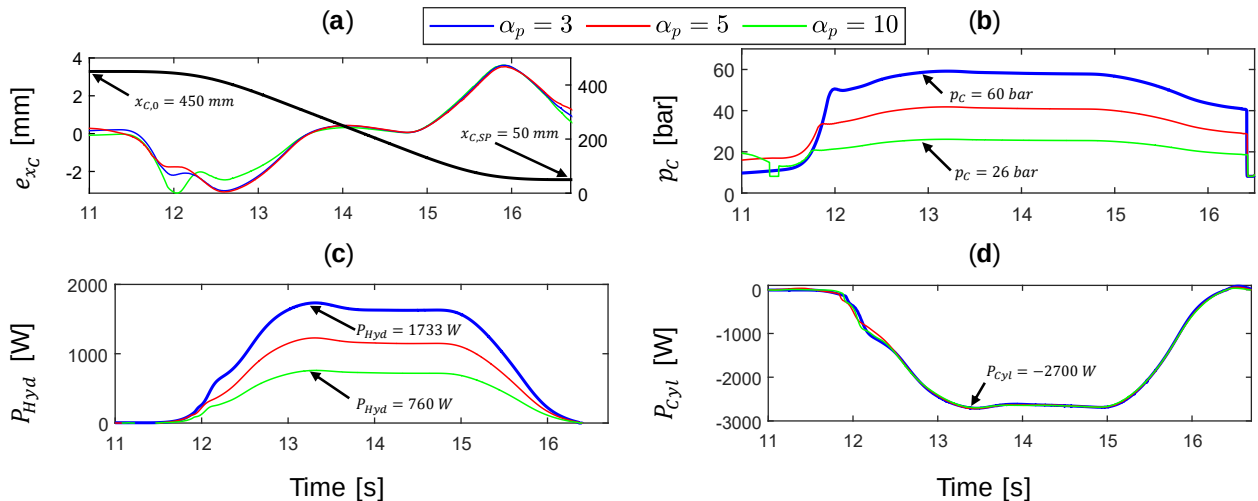


Figure 17. Simulated results with PF_{pL} and different pilot ratios of the OCV: (a) Position error; (b) Rod-side pressure; (c) Consumed hydraulic power; (d) Mechanical power at the actuator.

It can be concluded from Figure 17 (a) that varying the OCV's pilot ratio has little or no effect on the position tracking while the motion remains smooth. This is also confirmed by the trends of the pressure p_c sensed upstream the main spool of the PDCV that are depicted in Figure 17 (b); a lower value of this magnitude translates into energy savings since the required hydraulic power, defined via (27), drops down as visible in Figure 17 (c). In particular, increasing the pilot ratio up to $\alpha_p = 10$ reduces the power consumption to about 50% with respect to the case where $\alpha_p = 3$.

$$P_{Hyd} = p_c Q_B \quad (27)$$

The modifications that enable these energy savings do not affect the system performance because the mechanical power outputted by the actuator expressed in (28), remains about constant for every scenario (Figure 17 (d)).

$$P_{Cyl} = F_c v_c \quad (28)$$

6. CONCLUSION

The benefits of pressure feedback discussed in the technical literature are often demonstrated on simplified linear models, moreover these references do not show how the pressure feedback algorithm is implemented and tested on real applications using standard directional proportional control valves.

This paper investigates the implementation of pressure feedback on a hydraulically-actuated, single-boom crane operated in closed-loop position control. In comparison to conventional pressure feedback only based on the rod-side chamber pressure, the use of both actuator pressures achieves improved stabilization in both piston extension and retraction. To enable closed-loop stability without compromising the motion performance (i.e., settling time and tracking error), it is necessary to include the piston-side pressure for the considered application. However, adding the piston-side pressure to the pressure feedback induce instability due to the low bandwidth of the directional control valve. One solution presented in this paper is to account for the valve dynamics in the pressure feedback. The experimental results show relevant reduction of the oscillatory behavior of the actuator pressures in closed-loop motion control, the maximum position error during lifting of the payload is reduced from 3.5 mm to 1.0 mm, and the average position error when the payload is lowered is approximately 0.2 mm (in comparison to the system without pressure feedback, this is an improvement of almost 2 mm). Finally, this study demonstrates that the recommended pressure feedback allows the pilot-ratio of the over-center valve to be increased which results in a significant potential for energy savings.

Concerning the future work, the energy saving potential will be experimentally validated and the proposed pressure feedback will be tested on a hydraulic circuit with load-holding valves on both actuator ports in applications where the direction of the external load is changing (i.e., four quadrant operations).

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